

$$M(5, -6, 6)$$

$$\begin{array}{r|l} 0 & 27 \\ \hline 0 & 54 \\ -1 & 08 \\ 0 & 16 \\ 0 & 32 \\ 0 & 64 \\ -1 & 28 \\ -0 & 56 \end{array}$$

$$0.0100010 \mid 0$$

$$,10001:0 \mid -1$$

$$f(0,27) = [10001 \mid -1]$$

$$\varepsilon_0 = [100001 \mid -6] = \frac{1}{2} \cdot 2^{-6} = \underline{\underline{2^{-7}}}$$

$$M_{\infty} = [11111 \mid 6] = (1 - 2^{-5}) \cdot 2^6 = 2^6 - 2 = \underline{\underline{62}}$$

$$\varepsilon_1 = [10001 \mid 1] - [10000 \mid 1] = 2^{1-5} - 2^{-4} = \underline{\underline{2^{-4}}}$$

$$\begin{array}{r|l} 3 & 12 \\ \hline 1 & 1 \\ 0 & 1 \end{array} \quad \begin{array}{r|l} 8 \\ \hline 16 \\ 12 \\ 04 \\ 08 \\ 16 \end{array}$$

$$11.1100 \mid 0$$

$$.11110:0 \mid 2$$

$$f(3,8) = [11110 \mid 2]$$

$$\begin{array}{r} ,11110 \mid 2 \\ + ,10001 \mid -1 \\ \hline \end{array}$$

$$\begin{array}{r} ,11110 \mid 2 \\ + .00010:0 \mid 2 \\ \hline \end{array}$$

$$\begin{array}{r} 1.0000:0 \mid 2 \\ \hline [1.0000 \mid 3] \end{array}$$

$$\Delta f(0,27) = \frac{1}{2} \cdot 2^{-1-5} = \underline{\underline{2^{-7}}}$$

$$\Delta f(3,8) = \frac{1}{2} \cdot 2^{2-5} = \underline{\underline{2^{-4}}}$$

$$\Delta f(+) = \frac{1}{2} \cdot 2^{3-5} = \underline{\underline{2^{-3}}}$$

(2)

$$a = 6 \quad \Delta_a = \frac{5}{3}$$

$$b = 3 \quad \Delta_b = \frac{1}{2}$$

$$\frac{\sqrt{a \cdot b + b \cdot b}}{b}$$

$$x = \overset{18}{a \cdot b} + \overset{9}{b \cdot b} = 27$$

$$\Delta_x = 3 \cdot \Delta_a + 6 \cdot \Delta_b + 3 \cdot \Delta_b + 3 \Delta_b = 3 \cdot \frac{5}{3} + 12 \cdot \frac{1}{2} = 5 + 6 = 11$$

$$f(x) = x^{\frac{1}{2}} \rightarrow f(x) = \sqrt{x}$$

$$\Delta_{f(x)} = M_1 \cdot \Delta_x = \frac{1}{8} \cdot 11 = \frac{11}{8}$$

$$f'(x) = \frac{1}{2} \cdot x^{-\frac{1}{2}}$$

$$M_1 = \max_{\substack{[27-11, 27+11] \\ 16 \quad 38}} \left| \frac{1}{2} \cdot \frac{1}{\sqrt{x}} \right| = \frac{1}{2} \cdot \frac{1}{\sqrt{16}} = \frac{1}{8}$$

$$\Delta_{\sqrt{\frac{1}{b}}} = \frac{|b| \cdot \Delta_{f(x)} + \sqrt{27} \cdot \Delta_b}{b^2} = \frac{3 \cdot \frac{11}{8} + \sqrt{27} \cdot \frac{1}{2}}{9} = \frac{11}{24} + \frac{\sqrt{27}}{18}$$

$$\begin{bmatrix} 2 & -1 & 1 & -1 & | & -2 \\ -2 & -3 & 3 & 5 & | & -6 \\ -1 & -1 & 3 & 5 & | & -1 \\ -4 & -2 & 4 & 4 & | & -8 \end{bmatrix} \xrightarrow{\text{CSERE}} \begin{bmatrix} -4 & -2 & 4 & 4 & | & -8 \\ -2 & -3 & 3 & 5 & | & -6 \\ -1 & -1 & 3 & 5 & | & -1 \\ 2 & -1 & 1 & -1 & | & -2 \end{bmatrix} \begin{array}{l} -\frac{-2}{-4} = -\frac{1}{2} \\ -\frac{-1}{-4} = -\frac{1}{4} \\ -\frac{-2}{-4} = \frac{1}{2} \end{array}$$

$$\begin{bmatrix} -4 & -2 & 4 & 4 & | & -8 \\ 0 & -2 & 1 & 3 & | & -2 \\ 0 & -\frac{1}{2} & 2 & 4 & | & 1 \\ 0 & -2 & 3 & 1 & | & -6 \end{bmatrix} \xrightarrow{-\frac{1}{4}} \begin{bmatrix} -4 & -2 & 4 & 4 & | & -8 \\ 0 & -2 & 1 & 3 & | & -2 \\ 0 & 0 & \frac{7}{4} & \frac{13}{4} & | & \frac{3}{2} \\ 0 & 0 & 2 & -2 & | & -4 \end{bmatrix} \xrightarrow{\text{CSERE}}$$

$$\begin{bmatrix} -4 & -2 & 4 & 4 & | & -8 \\ 0 & -2 & 1 & 3 & | & -2 \\ 0 & 0 & 2 & -2 & | & -4 \\ 0 & 0 & \frac{7}{4} & \frac{13}{4} & | & \frac{3}{2} \end{bmatrix} \xrightarrow{-\frac{7/4}{2} = -\frac{7}{8}} \begin{bmatrix} -4 & -2 & 4 & 4 & | & -8 \\ 0 & -2 & 1 & 3 & | & -2 \\ 0 & 0 & 2 & -2 & | & -4 \\ 0 & 0 & 0 & 5 & | & 5 \end{bmatrix}$$

$$5x_4 = 5 \rightarrow x_4 = 1$$

$$2x_3 - 2 \cdot 1 = -4$$

$$x_3 = \frac{-2}{2} = -1$$

$$-2x_2 + 1 \cdot (-1) + 3 \cdot 1 = -2$$

$$x_2 = \frac{-4}{-2} = 2$$

$$-4x_1 - 2 \cdot 2 + 4 \cdot (-1) + 4 \cdot 1 = -8$$

$$x_1 = \frac{-4}{-4} = 1$$

$$\downarrow$$

$$\underline{\underline{x = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 1 \end{bmatrix}}}$$

$$\det(A) = -4 \cdot -2 \cdot 2 \cdot 5 \cdot (-1)^2 = \underline{\underline{80}}$$

(3)/2

$$\begin{bmatrix} \boxed{2} & -1 & 1 & -1 \\ -2 & -3 & 3 & 5 \\ -1 & -1 & 3 & 5 \\ -4 & -2 & 4 & 4 \end{bmatrix} \begin{array}{l} -\frac{-2}{2} = 1 \\ -\frac{-1}{2} = \frac{1}{2} \\ -\frac{-4}{2} = 2 \end{array} \rightarrow \begin{bmatrix} 2 & -1 & 1 & -1 \\ 0 & \boxed{-4} & 4 & 4 \\ 0 & -\frac{3}{2} & \frac{7}{2} & \frac{9}{2} \\ 0 & -4 & 6 & 2 \end{bmatrix} \begin{array}{l} \\ -\frac{-3/2}{-4} = -\frac{3}{8} \\ -\frac{-4}{-4} = -1 \end{array} \rightarrow$$

$$\rightarrow \begin{bmatrix} 2 & -1 & 1 & -1 \\ 0 & -4 & 4 & 4 \\ 0 & 0 & \boxed{2} & 3 \\ 0 & 0 & 2 & -2 \end{bmatrix} \begin{array}{l} \\ \\ -1 \end{array} \rightarrow \begin{bmatrix} 2 & -1 & 1 & -1 \\ 0 & -4 & 4 & 4 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & -5 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -\frac{1}{2} & \frac{3}{8} & 1 & 0 \\ -2 & 1 & 1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 4 & -2 & 2 \end{bmatrix} \quad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

(4)

$$\det(A) = 1 \cdot (2 - 4) + 4 \cdot (-4) = -18$$

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)^T = \frac{1}{-18} \cdot \begin{bmatrix} -2 & -8 & -4 \\ -8 & -14 & 2 \\ -4 & 2 & 1 \end{bmatrix}^T = -\frac{1}{18} \cdot \begin{bmatrix} -2 & -8 & -4 \\ -8 & -14 & 2 \\ -4 & 2 & 1 \end{bmatrix}$$

$$\text{cond}_{\infty}(A) = \|A\|_{\infty} \cdot \|A^{-1}\|_{\infty} = 8 \cdot \frac{1}{18} \cdot 8 = \frac{32}{3}$$

$$\text{cond}_2(A) = \frac{\max |\lambda_i(A)|}{\min |\lambda_i(A)|} = \frac{6}{1} = 6$$

$$\nearrow \\ A = A^T$$

$$\begin{bmatrix} 1-\lambda & 0 & 4 \\ 0 & 1-\lambda & -2 \\ 4 & -2 & 2-\lambda \end{bmatrix}$$

$$\begin{aligned} & (1-\lambda) \cdot \left[(1-\lambda) \cdot (2-\lambda) - 4 \right] + \\ & + 4 \cdot \left[0 - 4 \cdot (1-\lambda) \right] = \\ & = (1-\lambda) \cdot \left[(1-\lambda) \cdot (2-\lambda) - 4 - 16 \right] = \\ & = (1-\lambda) \cdot \left[\lambda^2 - 3\lambda - 18 \right] \end{aligned}$$

$$\lambda_1 = 1$$

$$\lambda_{2,3} = \frac{3 \pm \sqrt{9 + 4 \cdot 18}}{2} = \begin{cases} 6 = \lambda_2 \\ -3 = \lambda_3 \end{cases}$$

$$x^4 - 4x^3 - x^2 + 16x - 12$$

5

	1	-4	-1	16	-12
ξ	2	$1 \cdot 2 = 2$	$-2 \cdot 2 = -4$	$-5 \cdot 2 = -10$	$6 \cdot 2 = 12$
$a_i^{(1)}$	1	$2 - 4 = -2$	$-4 - 1 = -5$	$-10 + 16 = 6$	$12 - 12 = \boxed{0}$
ξ	2	$1 \cdot 2 = 2$	0	$-5 \cdot 2 = -10$	
$a_i^{(2)}$	1	$2 + 2 = \cancel{4}^0$	-5	$-10 + 6 = \boxed{-4}$	
ξ	2	$1 \cdot 2 = 2$	$2 \cdot 2 = 4$		
$a_i^{(3)}$	1	$2 + 0 = 2$	$4 - 5 = \boxed{-1}$		
ξ	2	$1 \cdot 2 = 2$			
$a_i^{(4)}$	1	$2 + 2 = \boxed{4}$			
ξ	2				
$a_i^{(5)}$	$\boxed{1}$				

$$P(x) = 0 - 4 \cdot (x-2) - 1 \cdot (x-2)^2 + 4 \cdot (x-2)^3 + (x-2)^4$$

$$r < |x_2| < R$$

$$r = \frac{1}{1 + \frac{\max_{i=1}^{n-1} |a_i|}{|a_n|}} = \frac{1}{1 + \frac{16}{12}} = \frac{12}{28} = \frac{3}{7}$$

$$R = 1 + \frac{\max_{i=0}^{n-1} |a_i|}{|a_n|} = 1 + \frac{16}{1} = 17$$